

ORB Strategy V3 - Experiment Tracker & Continuation Guide

PROJECT OVERVIEW

What We're Doing

Developing and optimizing an Opening Range Breakout (ORB) strategy for NQ/MNQ futures trading using Pine Script on TradingView. The strategy is based on ICT (Inner Circle Trader) methodology and incorporates multiple filters and market structure concepts.

Current Strategy Versions

- **V2:** Simpler version (488 lines), focused 9:31-10:00 AM window, stop after win, max 3 attempts
- **V3:** Complex version (1,300+ lines), all-day trading, market structure levels, confluence scoring
- **V3 Momentum:** V3 + Premium/Discount momentum filter (1,455 lines)

Key Files & Locations

Pine Script Files (TradingView)

File	Location	Description
V2	<code>/mnt/user-data/outputs/orb_strategy_v2.pine</code>	Simple, focused strategy
V3	<code>/mnt/user-data/outputs/orb_v3_market_structure.pine</code>	Full market structure version
V3 Momentum	<code>/mnt/user-data/outputs/orb_v3_momentum.pine</code>	V3 + momentum filter

Data Files

File	Location	Description
Raw trades with context	/home/claude/trades_with_all_context.pkl	1,685 trades with PDH/PDL/ONH/ONL context
Analysis data	/home/claude/trades_with_context.pkl	1,648 trades for analysis

Backtest Results (Uploaded by User)

File	Test Configuration
ORB_V3_MS_Mom_CME_MINI_MNQ1_2026-01-06_36b8c.xlsx	Baseline (Momentum OFF)
ORB_V3_MS_Mom_CME_MINI_MNQ1_2026-01-06_e92f2.xlsx	Extended + Previous Day
ORB_V3_MS_Mom_CME_MINI_MNQ1_2026-01-06_9525e.xlsx	Extended + Overnight
ORB_V3_MS_Mom_CME_MINI_MNQ1_2026-01-06_845b0.xlsx	Value + Previous Day



COMPLETED EXPERIMENTS

Experiment 1: Premium/Discount Zone Analysis (Python)

Date: 2026-01-06
Data: 1,685 trades from TradingView backtest export
Objective: Determine if filtering trades based on Premium/Discount zones improves win rate
Methodology:

- Calculated 4 zone definitions: Previous Day (PDH/PDL), Overnight (ONH/ONL), Session Range, Previous Hour
- Tested 2 modes: "Extended" (Long in Premium, Short in Discount) vs "Value" (traditional ICT)
- Analyzed win rate impact

Python Analysis Results (from raw trade data):

Zone Reference	Extended WR	Value WR	Difference
Previous Day	42.3%	19.0%	+23.3%
Overnight	41.1%	17.8%	+23.3%
Session Range	38.5%	22.1%	+16.4%
Previous Hour	31.8%	29.9%	+1.9%

Initial Conclusion: Extended mode appeared significantly better in raw data analysis.

Experiment 2: V2 vs V3 Comparison Analysis

Date: 2026-01-06

Objective: Understand why V2 performs better than V3 in practice

Key Differences Identified:

Feature	V2	V3
Trading Window	9:31-10:00 AM (29 min)	9:31-15:45 (all day)
Max Attempts	3	15
Stop After Win	YES (always)	Optional (OFF)
Entry Mode	Pullback + Fallback	Immediate
TP Target	0.10%	0.25%
MAE Threshold	0.25%	0.10%
Complexity	488 lines	1,300+ lines

Hypotheses for V2 Superiority:

1. Narrow trading window captures best 9-10 AM edge
2. Stop after win prevents overtrading
3. Fewer attempts = higher quality trades
4. Pullback entries filter false breakouts
5. Quick TP (0.10%) = higher hit rate
6. Higher MAE tolerance gives trades room to work

Experiment 3: V3 Momentum Filter Live Backtest

Date: 2026-01-06

Objective: Validate Python analysis findings in actual TradingView backtest

Test Configurations:

1. Baseline: Momentum Filter OFF
2. Extended + Previous Day: ON, Previous Day, Extended, Skip Trade
3. Extended + Overnight: ON, Overnight, Extended, Skip Trade

4. Value + Previous Day: ON, Previous Day, Value, Skip Trade

ACTUAL BACKTEST RESULTS (TradingView):

Test	Trades	Win Rate	Net P&L	Avg Trade	PF	Max DD
Baseline (OFF)	2,269	50.99%	\$3,812	\$1.68	1.09	\$1,263
Extended + Overnight	1,580	50.63%	\$1,984	\$1.26	1.06	\$1,691
Extended + Previous Day	1,395	50.25%	\$1,209	\$0.87	1.04	\$1,600
Value + Previous Day	1,203	52.20%	\$3,306	\$2.75	1.15	\$792

 CRITICAL FINDING: Python Analysis vs Live Backtest Discrepancy

Python analysis predicted: Extended mode wins (+23% WR improvement)

Live backtest showed: VALUE mode wins!

Why the discrepancy?

1. Multi-TP exits change P&L dynamics (Python assumed simple TP hit/miss)
2. MAE filter, breakeven moves, and other V3 filters interact with momentum filter
3. "Extended" filtered OUT reversal setups that were actually GOOD in V3's implementation

 VALIDATED BEST SETTINGS (V3 Momentum):

Setting	Value
Enable Momentum Filter	ON
Zone Reference	Previous Day
Momentum Mode	VALUE
Filter Action	Skip Trade

Benefits of VALUE mode:

- +1.21% higher win rate (52.20% vs 50.99%)
 - +64% better avg trade (\$2.75 vs \$1.68)
 - +5.5% better profit factor (1.15 vs 1.09)
 - 37% lower max drawdown (\$792 vs \$1,263)
 - 47% fewer trades with nearly same profit
-

PENDING EXPERIMENTS

To-Do List

1. Test V3 with V2-like settings:

- Change trading window to 9:31-10:00 AM
- Enable Stop After Win
- Max Attempts = 3
- Entry Mode = Pullback + Fallback

2. Add Momentum Filter to V2:

- Keep V2's simplicity
- Add only the momentum filter (Value mode, Previous Day)

3. Create V4 "Best of Both":

- V2 core logic + V3 market structure + Momentum Filter (Value mode)

4. Hour-by-Hour Analysis:

- Validate which hours are profitable in V3 backtest data
- Previous analysis showed 9-10 AM best, 1-3 PM negative

5. Confluence Score Analysis:

- Analyze trades by confluence score
- Determine optimal minimum score threshold

UTILITY SCRIPTS & CODE SNIPPETS

Reading TradingView Excel Exports

```
python
```

```

import pandas as pd

def analyze_tv_backtest(filepath):
    """Read TradingView strategy export and extract key metrics"""
    xl = pd.ExcelFile(filepath)

    # Get settings from Properties sheet
    props = pd.read_excel(filepath, sheet_name='Properties')
    props = props.set_index('name')

    # Get trade analysis
    trades_analysis = pd.read_excel(filepath, sheet_name='Trades analysis')
    trades_analysis = trades_analysis.set_index('Unnamed: 0')

    # Get performance summary
    perf = pd.read_excel(filepath, sheet_name='Performance')
    perf = perf.set_index('Unnamed: 0')

    # Get list of trades
    trades = pd.read_excel(filepath, sheet_name='List of trades')

    # Extract key metrics
    total_trades = trades_analysis.loc['Total trades', 'All USD']
    win_rate = trades_analysis.loc['Percent profitable', 'All %']
    avg_pnl = trades_analysis.loc['Avg P&L', 'All USD']
    profit_factor = pd.read_excel(filepath, sheet_name='Risk-adjusted performance')
    net_profit = perf.loc['Net profit', 'All USD']
    max_dd = perf.loc['Max equity drawdown (intrabar)', 'All USD']

    # Key settings
    momentum_enabled = props.loc['Enable Momentum Filter', 'value']
    zone_ref = props.loc['Zone Reference', 'value']
    momentum_mode = props.loc['Momentum Mode', 'value']

    return {
        'total_trades': total_trades,
        'win_rate': win_rate,
        'avg_pnl': avg_pnl,
        'profit_factor': profit_factor,
        'net_profit': net_profit,
        'max_dd': max_dd,
    }

```



```
'momentum_enabled': momentum_enabled,  
'zone_ref': zone_ref,  
'momentum_mode': momentum_mode  
}
```

Loading Pickle Data

```
python  
  
import pandas as pd  
  
# Load trades with context  
df = pd.read_pickle('/home/claude/trades_with_all_context.pkl')  
  
# Key columns: entry_time, exit_time, direction, pnl, pdh, pdl, onh, onl, entry
```



CONTINUATION PROMPT

Use this prompt to resume work:

PROMPT TO CONTINUE:

I'm continuing work on my ORB strategy optimization. Please read the experiment tracker document I'm sharing to understand where we left off.

- Key context:
- We're optimizing an ORB (Opening Range Breakout) strategy for NQ/MNQ futures
 - We have V2 (simple), V3 (complex), and V3 Momentum versions
 - Latest finding: VALUE mode momentum filter (traditional ICT) outperforms Extended mode in live backtest
 - Best settings: Momentum ON, Previous Day zone, VALUE mode, Skip Trade action

- Files you may need:
- Pine scripts in /mnt/user-data/outputs/
 - Trade data in /home/claude/ (pickle files)
 - Backtest results uploaded as Excel files

What I want to do next: [SPECIFY YOUR NEXT TASK]

- Options we discussed:
1. Test V3 with V2-like settings (narrow window, stop after win)
 2. Add momentum filter to V2
 3. Create V4 combining best of both
 4. Analyze trades by hour or confluence score

 QUICK REFERENCE: V3 Momentum Filter Settings

Zone Reference Options

Option	Description	Effectiveness
Previous Day	PDH/PDL midpoint	★ BEST
Overnight	ONH/ONL midpoint	Good
Session Range	Today's expanding H/L	Moderate
Previous Hour	Last hour's H/L	Weak

Momentum Mode Options

Option	Logic	Result
Extended	Long in Premium, Short in Discount	✖ Underperformed
Value	Long in Discount, Short in Premium	✔ BEST
Extended Only Longs	Filter only applies to longs	Not tested
Extended Only Shorts	Filter only applies to shorts	Not tested

Filter Action Options

Option	Behavior
Skip Trade	Don't enter if filter fails
Reduce Size	Enter with 50% position
Warn Only	Enter anyway, just track

 CHANGE LOG

Date	Change	Result
2026-01-06	Created V3 Momentum filter	Added Premium/Discount filtering to V3
2026-01-06	Ran 4 backtest configurations	VALUE mode outperformed all others
2026-01-06	Analyzed V2 vs V3 differences	Identified 7 key factors for V2 superiority

 EXPERIMENT 4: COMPREHENSIVE FAILURE ANALYSIS

Date: 2026-01-06
Data: 1,685 trades + 1-min OHLC data
Objective: Deep dive into why 69.7% of trades fail - identify patterns and actionable filters

Key Findings

1. PROXIMITY TO KEY LEVELS AT ENTRY

Level	% of Losers Close (<0.1%)	Insight
ON_Mid	39.1%	⚠️ HIGHEST - Equilibrium is rejection zone!
PD_Mid	17.0%	Also dangerous
PDH	16.3%	Key resistance
PDL	10.7%	Key support
ONH	4.8%	Less frequent
ONL	0.3%	Rarely hit

INSIGHT: ~40% of losers entered very close to equilibrium levels (midpoints). These are REJECTION zones, not entry zones!

2. WHERE DID PRICE REVERSE? (MFE Analysis)

- 42.4% of losers NEVER went into profit (MFE ≈ 0)
- 55.2% reversed VERY CLOSE to a key level (<0.1%)

Top Rejection Levels (where MFE peaked before reversal):

Level	% of Reversals
ONH	26.4%
ON_Mid	22.7%
ONL	17.6%
PDL	13.2%
PD_Mid	12.4%
PDH	7.7%

INSIGHT: Overnight levels (ONH/ONL/ON_Mid) are the PRIMARY rejection zones!

3. TRADING AGAINST MARKET STRUCTURE (15-min HH/HL/LH/LL)

Scenario	Count	% of Losers
Long in BEARISH structure	155	28.8% of long losers
Short in BULLISH structure	226	35.5% of short losers
Trading in RANGING structure	472	40.2% of all losers

INSIGHT: ~30% of losses are from trading AGAINST the HTF structure!

4. HTF FVG ANALYSIS

1-Hour FVGs:

- 10.3% of losers entered INSIDE a bullish FVG
- 8.3% entered INSIDE a bearish FVG
- 71.9% entered NEAR an FVG (within 0.2%)

Daily FVGs:

- Only 6.2% of losers were near a Daily FVG

- Daily FVGs are less frequent but more significant

5. TIME-BASED ANALYSIS (Killzones)

Best Entry Windows:

Time	Win Rate	Notes
09:39	38.1%	Best!
09:32	33.8%	First entries
09:33	33.6%	Still good

Worst Entry Windows:

Time	Win Rate	Notes
09:42	16.7%	Avoid!
09:43	9.1%	WORST!
10:01	22.7%	Avoid

INSIGHT: First 2-3 minutes (09:32-09:34) work best. 09:42-09:43 is a death zone!

6. PREMIUM/DISCOUNT ZONE FAILURES

Long Losers:

- 45.9% in Premium (going Long in expensive zone = BAD)
- 51.3% in Discount (should be good, but still losing)

Short Losers:

- 62.0% in Premium (should be good, but still losing)
- 36.1% in Discount (shorting cheap zone = BAD)

INSIGHT: Zone alone isn't enough - need confluence with other factors!

RECOMMENDED FILTERS (Priority Order)

1. EQUILIBRIUM FILTER (HIGHEST IMPACT)

- Skip trades within 0.1% of ON_Mid or PD_Mid
- These are rejection zones, not entry zones
- **Expected impact: Filter ~40% of losers**

2. TIME WINDOW FILTER (EASY TO IMPLEMENT)

- Best entries: 09:32-09:39
- Avoid or reduce size: 09:42-09:43, 10:01
- Consider only first 10 minutes after ORB formation

3. HTF STRUCTURE FILTER (MEDIUM COMPLEXITY)

- Only Long in bullish structure (HH/HL on 15-min)
- Only Short in bearish structure (LH/LL on 15-min)
- Skip ranging market or trade both directions with caution
- **Expected impact: Filter ~30% of losers**

4. OVERNIGHT LEVEL AWARENESS (TP/SL OPTIMIZATION)

- Track ONH/ONL/ON_Mid as key rejection levels
- For Longs: Avoid if ONH is very close above entry
- For Shorts: Avoid if ONL is very close below entry

5. HTF FVG CONFLUENCE (ADVANCED)

- For Longs: Prefer entries inside/near bullish 1H FVG
- For Shorts: Prefer entries inside/near bearish 1H FVG

- Daily FVG = highest conviction

ICT CONCEPTS TO EXPLORE FURTHER

Based on failure analysis, these ICT concepts warrant investigation:

1. **CISD (Change in State of Delivery)** - Market structure shifts
2. **MSS (Market Structure Shift)** - Break of structure confirmation
3. **Order Blocks** - 1H OBs showed only 7.2% of losers inside them (may need refinement)
4. **Breaker Blocks** - Not yet analyzed
5. **Liquidity Pools** - Above/below swing highs/lows
6. **Midnight Open** - 24.8% of losers within 0.1% (significant reference point)
7. **NY Midnight/True Day Open** - Key ICT time reference
8. **Weekly Open** - Not yet analyzed
9. **Previous Week High/Low** - Not yet analyzed

Updated: 2026-01-06

EXPERIMENT 5: V2 COMPARISON & DEEP ICT ANALYSIS

Date: 2026-01-06

Data: V2 backtest (1,019 trades) + V3 data (1,685 trades) + 1-min OHLC

Objective: Compare V2 vs V3 failure patterns, explore deeper ICT concepts

V2 Performance Summary

Metric	V2	V3 Baseline	V3 + Value Mom
Trades	1,019	2,269	1,203
Win Rate	44.3%	51.0%	52.2%
Avg Trade	\$5.24	\$1.68	\$2.75
Profit Factor	1.26	1.09	1.15
Net Profit	\$5,335	\$3,812	\$3,306
Max Drawdown	\$613	\$1,263	\$792

V2 Key Settings: Window 09:31-10:00, Max 2 attempts, MAE 0.15%

V2 vs V3 Failure Pattern Comparison

Level	V3 Losers	V2 Losers	Difference
ON_Mid	39.1%	30.0%	V2 9.1% better
PD_Mid	17.0%	13.0%	V2 4.0% better
PDH	16.3%	18.3%	Similar
ONH	4.8%	14.7%	V2 worse (more ONH entries)

Key Insight: V2's narrow window naturally avoids equilibrium entries!

Time-Based Death Zone (Both Strategies)

Time	V2 WR	V3 WR	Status
09:32	47.7%	33.8%	✔ Good
09:43	31.2%	9.1%	✖ DEATH ZONE
09:44	20.0%	-	✖ DEATH ZONE

Time	V2 WR	V3 WR	Status
09:46	28.6%	-	✗ DEATH ZONE

Cause: This time window coincides with LIQUIDITY SWEEPS on initial range.

🔥 MAJOR ICT DISCOVERIES

Discovery 1: GAP DIRECTION FILTER (BIGGEST EDGE!)

Scenario	Trades	Win Rate	Impact
Long on Gap UP	371	45.3%	✓✓ BEST
Long on Gap DOWN	246	17.1%	✗✗ WORST
Short on Gap UP	381	15.5%	✗✗ WORST
Short on Gap DOWN	318	42.5%	✓✓ BEST

⚡ **RULE:** Trade WITH gap direction = 43-45% WR. Against gap = 15-17% WR.

This is a ~28% DIFFERENCE - the single biggest filter we've found!

Discovery 2: TRUE DAY OPEN (MIDNIGHT) FILTER

Scenario	Trades	Win Rate	Improvement
Long ABOVE TDO	489	37.6%	+11% vs below
Long BELOW TDO	307	26.4%	Avoid
Short ABOVE TDO	478	21.5%	Avoid
Short BELOW TDO	393	34.6%	+13% vs above

⚡ **RULE:** Long only above TDO, Short only below TDO.

Discovery 3: Combined Filter Projection

If we apply Gap Direction + TDO filters:

- Long: Gap UP + Above TDO → **potentially 50%+ WR**
- Short: Gap DOWN + Below TDO → **potentially 50%+ WR**

 RECOMMENDED FILTER STACK (Priority Order)

#	Filter	Logic	Expected Impact
1	GAP DIRECTION	Long only Gap UP, Short only Gap DOWN	+28% WR
2	TRUE DAY OPEN	Long above TDO, Short below TDO	+11-13% WR
3	EQUILIBRIUM AVOIDANCE	Skip within 0.1% of ON_Mid/PD_Mid	Filter ~40% losers
4	TIME WINDOW	Best 09:32-09:39, Skip 09:42-09:46	Avoid death zone
5	VALUE MODE MOMENTUM	Long in Discount, Short in Premium	+1.2% WR (validated)
6	OVERNIGHT LEVELS	TP before ONH/ONL/ON_Mid	Better exits

 ICT CONCEPTS ANALYZED

Concept	Finding	Actionable?
Gap Direction	+28% WR when aligned	 HIGH PRIORITY
True Day Open	+11-13% WR when aligned	 HIGH PRIORITY

Concept	Finding	Actionable?
Equilibrium Zones	39% losers entered near	✅ Avoid midpoints
Liquidity Sweeps	09:42-09:46 death zone	✅ Skip this window
MSS (Market Structure Shift)	Detected on 5-min	🔍 Needs more analysis
1H FVGs	18.6% losers inside FVG	🔍 Moderate signal
Daily FVGs	Only 6.2% near	🔍 Rare but significant
Order Blocks	7.2% inside 1H OB	🔍 Needs refinement
Weekly Levels	PWH/PWL/Weekly Open	🔍 Some signal
Breaker Blocks	Not analyzed	📝 TODO



NEXT IMPLEMENTATION STEPS

1. Add **GAP DIRECTION** filter to **V3** (Highest impact)
2. Add **TRUE DAY OPEN** filter to **V3**
3. Test combined filter stack in backtest
4. Consider creating **V4** with all filters

Updated: 2026-01-06